

Phys 20AL Week 8: Path of Light

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1 Aim for the Experiment

The aim for the experiment is to study the path that light travels: First it'll be focused on using Snell's Law to calculate a half-cylinder's index of refraction (that's independent of the frequency of the light). Later on it'll be focused on using the prism to get a frequency-dependent index of refraction for a prism instead.

2 Experimental Setup

The equipments include: A light source (with both white light, and RGB lights), a protractor, and transparent half-cylinder and triangular prism.

The equipment setup is also simple: setup the light so the path of the light passes through where the half-cylinder and the triangular prism are going to be at, and put a protractor below.



Figure 1: The actual lab setup picture

3 Quantity to Measure & Method of Measure

For measuring the index of refraction through Snell's Law, the incident angle of the light away from the normal line of the surface will be measured (for both the input and output light), and it'll be measured

through the protractor.

For both experiments (using the white light and the RGB light), there are total of 5 incident angles: $10^\circ, 20^\circ, 30^\circ, 40^\circ, 50^\circ$ that are used as manipulated variable. For the second experiment specifically (the RGB light one), the color (or frequency) of the light is also a manipulated variable.

For each pair of manipulated variables, total of 3 trials for measurement will be performed.

4 Experimental Procedure

4.1 Frequency Independent Experiment

It'll follow the procedure below:

1. Turn on the white light, and put the half-cylinder on the path of the light, with the flat side facing toward the light.
2. Tweak the cylinder, so the light is passing directly through the center of the half-cylinder (or, the center point of the semicircle).
3. Adjust the angle of the cylinder, so the light has incident angle as one of the angles to be measured. Then, measure the incident angle of the output light.
4. Repeat Step 3 for all the other angles await to be measured.
5. Repeat Step 3-4 for a total of 3 trials to get all the desired measurements.

Afterward, using the gathered data to calculate the index of refraction of the half-cylinder, use it to predict the critical angle, and test how well the prediction is experimentally.

4.2 Frequency Dependent Experiment

In this case, the experimental procedure is quite similar to the previous one, except now there are 3 different light sources - red, green, and blue. So, the experiment described above needs to be repeated for all types of light source, to get the index of refraction with respect to each frequency of light.

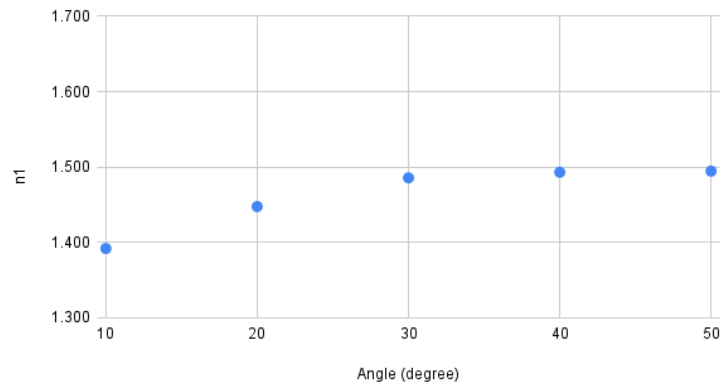
5 Experimental Data

The experimental data for each is as follow:

5.1 Frequency Independent Experiment

1	Input Angle (deg) \ Trial	1	2	3 Ave	SD	n1	
2	10	7.500	7.000	7.000	7.167	0.289	1.392
3	20	14.000	13.500	13.500	13.667	0.289	1.448
4	30	19.500	19.500	20.000	19.667	0.289	1.486
5	40	25.500	25.500	25.500	25.500	0.000	1.493
6	50	31.000	31.000	30.500	30.833	0.289	1.495
7							
8					Ave n1		1.463
9							
10					critical angle:		43.136
11							

n1 vs. Angle (degree)



5.2 Frequency Dependent Experiment

1	Red light:								
2	Input Angle (deg)\Trial	1	2	3 Ave	SD	n1		Ave n1	
3	10	22.500	20.000	19.500	20.667	1.607	0.695	1.556	
4	20	37.000	38.000	37.500	37.500	0.500	1.251		
5	30	50.000	49.500	50.000	49.833	0.289	1.658		
6	40	57.500	59.000	58.500	58.333	0.764	1.955		
7	50	68.500	68.500	69.500	68.833	0.577	2.221		
8									
9									
10	Green light:								
11	Input Angle (deg)\Trial	1	2	3 Ave	SD	n1		Ave n1	
12	10	27.000	26.000	26.000	26.333	0.577	0.820	1.614	
13	20	40.000	41.500	42.000	41.167	1.041	1.318		
14	30	51.500	51.500	53.000	52.000	0.866	1.690		
15	40	63.000	62.500	63.500	63.000	0.500	2.008		
16	50	71.500	70.000	70.500	70.667	0.764	2.236		
17									
18									
19	Blue light:								
20	Input Angle (deg)\Trial	1	2	3 Ave	SD	n1		Ave n1	
21	10	27.000	26.000	25.000	26.000	1.000	0.812	1.615	
22	20	41.000	40.000	41.000	40.667	0.577	1.309		
23	30	53.000	52.500	51.500	52.333	0.764	1.695		
24	40	65.000	63.000	63.500	63.833	1.041	2.017		
25	50	72.000	71.000	71.500	71.500	0.500	2.242		

n1 vs. Light Source

